

01

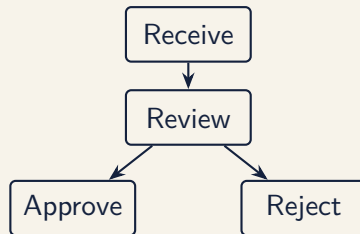
Process Thinking and Process Mining

Start with the operational question, not the algorithm.

A process is behavior over time

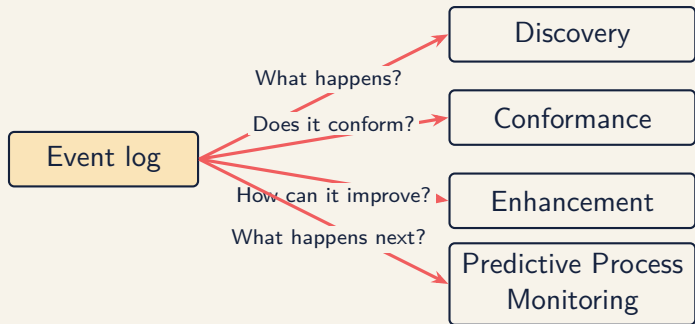
A process consists of:

- ▶ **Cases:** process instances
- ▶ **Activities:** meaningful steps
- ▶ **Events:** recorded activity executions
- ▶ **Resources:** people or systems involved
- ▶ **Outcomes:** time, cost, quality, compliance



Key idea A process model is a claim about possible behavior. An event log is evidence of observed behavior.

Four questions, four branches



Discover

Reveal actual behavior.

Check

Compare log and model.

Enhance

Add time, cost, or decisions.

Predict

Anticipate outcomes, delays, and risks.

Process mining is not a dashboard

	Typical analytics	Process mining
Unit	Rows and aggregates	Cases and ordered events
Question	How many?	How did work unfold?
Structure	Usually tabular	Sequential, concurrent, cyclic
Output	KPI or prediction	Behavior, deviation, performance
Risk	Missed context	Misleading case/event semantics

Discuss A delivery KPI worsened. What sequence-level questions would you ask before recommending a change?

Frame a useful analysis

1. **Decision:** What decision should this analysis support?
2. **Scope:** Where does the process begin and end?
3. **Case:** What is one process instance?
4. **Evidence:** Which systems record relevant events?
5. **Success:** Which behavior or outcome should change?

Example

“Why are some orders late?” becomes: “Which paths, handovers, and waiting periods distinguish orders delivered after the five-day target?”

02

Event Data and Event-Log Semantics

The quality of the analysis begins with the meaning of an event.

The minimum viable event log

Case	Activity	Timestamp	Resource	Amount
O-17	Receive order	09:02	Portal	480
O-17	Check credit	09:16	Elena	480
O-18	Receive order	09:12	Portal	125
O-17	Approve order	10:03	Marco	480
O-18	Check credit	10:24	Elena	125

Case ID

Groups related events.

Activity

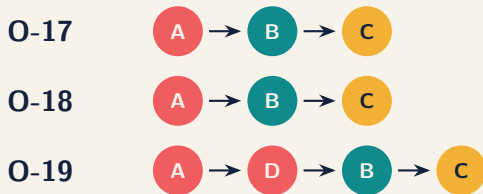
Names what happened.

Timestamp

Orders events in time.

Key idea Resources, costs, lifecycle states, and business attributes enrich analysis, but they do not repair an incoherent case notion.

From events to traces and variants



Trace: the ordered activity sequence for one case.

Variant: a distinct trace pattern shared by cases.

Here: 3 cases, 2 variants.

Lifecycle semantics change the answer



One activity may emit several events

- ▶ Start
- ▶ Suspend / resume
- ▶ Complete

Key idea Without lifecycle information, elapsed time may be mistaken for active work time.

A first PM4Py event log

```
import pandas as pd
import pm4py

events = pd.read_csv("orders.csv")
events["timestamp"] = pd.to_datetime(
    events["timestamp"], utc=True
)

events = pm4py.format_dataframe(
    events,
    case_id="case_id",
    activity_key="activity",
    timestamp_key="timestamp",
)

pm4py.write_xes(events, "orders.xes")
```

Before running discovery: verify identifiers, timestamps, ordering, duplicates, and event meaning.